AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

Mathematics Paper II

Time: 2 hours 15 minutes Marks: 65

AKU May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 5 Marks)

i. Evaluate $\lim_{x\to 9}\frac{x^2-81}{\sqrt{x}-3}$. (3 Marks)

ii. If $f(x)=4-x$ and $g(x)=\frac{9}{x+2}$, then find $f(g(x))$ and its domain. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

Q.2. (Total 12 Marks)

a. A curve has the equation $y = \frac{12}{x^2 - 4x}$. Find

i. $\frac{dy}{dx}$. (2 Marks)

ii. the equation of the normal to the curve y at $x = 3$. (4 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

b. Find the derivative of the following with respect to x.

i. $y = \ln(\sin x + \cos x)$ (2 Marks)

ii. $y = \dfrac{x + e^{3x}}{\sqrt{x}}$ (4 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

c. For the function $y = 50x^2 - \frac{9}{2}x^4$, find the values of x for which the function y has extreme values. Apply second derivative rule to find the minimum value of the given function. (6 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

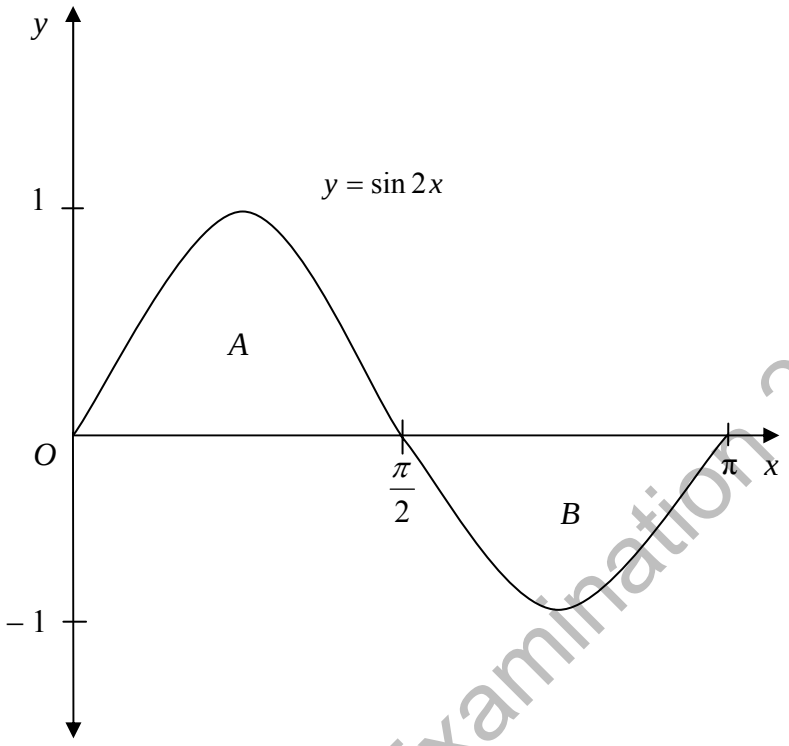
Q.3. (Total 12 Marks)

a. Resolve $\frac{3x^2+5}{(x+1)^2(x-1)}$ into its partial fractions. (6 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

b. The given figure shows the region *A* and *B* enclosed by the curve $y = \sin 2x$ and *x*-axis.



i. Find the area of region *A*. (5 Marks)

ii. Hence, find the area of region *B*. (1 Mark)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

c. Evaluate $\int x^2 \cos x \, dx$. (6 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

Q.4. (Total 10 Marks)

a. For the lines $l_1 : y = -5x + 7$ and $l_2 : y = 3x - 8$, find the

i. angle from l_1 to l_2. (3 Marks)

ii. equation of a line that is parallel to l_2 and passes through the point $(2, -3)$. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

b.

i. Find the perpendicular distance of a point $(5, -7)$ to the line $2x - 3y - 13 = 0$. (3 Marks)

ii. Find the equation of a line that is perpendicular to the line $y = 5x + 9$ and cuts y-axis at 3. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

c. Find equation of perpendicular bisector of the line joining the points (– 3, 10) and (– 5, 14). This perpendicular bisector meets *x*-axis at a point *Q*. Find the co-ordinates of the point *Q*. (5 Marks)

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Q.5. (Total 6 Marks)

A car manufacturing company manufactures 800 CC and 1000 CC cars in two factories *A* and *B*. The number of cars manufactured daily in each factory is shown in the given table.

	800 CC	**1000 CC**
Factory *A*	10	15
Factory *B*	8	12

The daily cost of operating the factories *A* and *B* is Rs 3,000,000 and Rs 2,200,000 respectively. Factories *A* and *B* work together to complete an order of 50 cars of 800 CC and 80 cars of 1000 CC. If *x* and *y* are the number of days factories *A* and *B* took to complete the order, find the constraints and cost function to complete this order.

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 6 Marks)

a. The line $x - y = 2$ is tangent to a circle at point (6, 4). The point (8, 0) lies on this circle. Find the centre of this circle.

b. Find equation of the tangent to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ which is parallel to the line $3x - 4y - 1 = 0$.

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

Q.7. (Total 10 Marks)

a. Find the equations of tangents to the parabola $y^2 = 8x$ at the points where $x = 2$. (5 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

b. Find the coordinates of centre, vertices, covertices and foci of the ellipse $\frac{(x+4)^2}{4}+\frac{y^2}{9}=1$.

(5 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

c. Find the equation of hyperbola that has its centre at origin. Its eccentricity is $\frac{\sqrt{65}}{4}$ and its directrices are $y = \pm \frac{16}{\sqrt{65}}$. (5 Marks)

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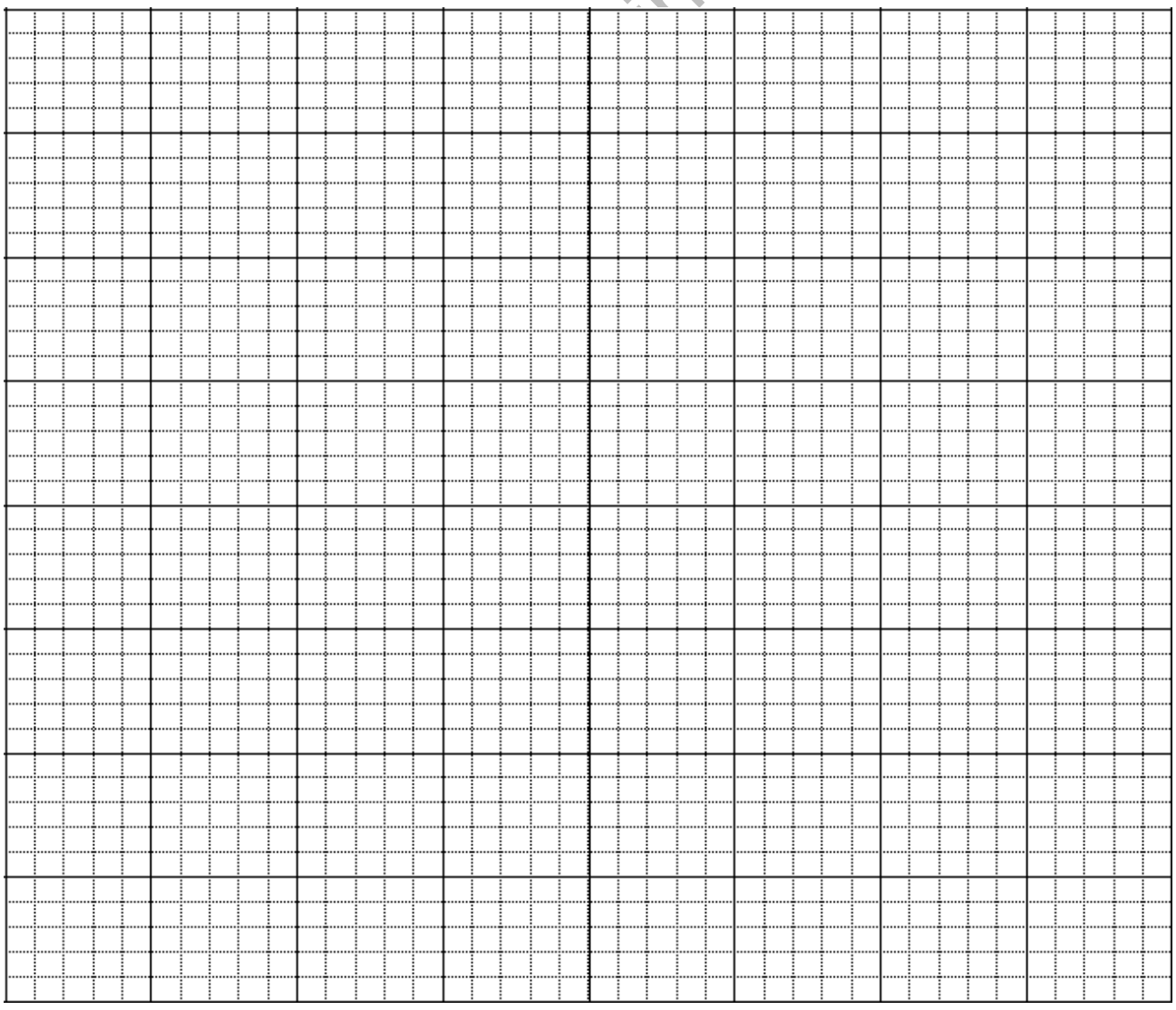
(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 4 Marks)

a. A triangle *ABC* located on *xy*-coordinate axes has vertices *A* (0, 3), *B* (– 3, 0) and *C* (3, – 3). The *x*-axis is translated through a distance of – 3 units with respect to origin while there is no change in the *y*-axis.

i. Find the new vertices of the triangle *ABC* with respect to the translated axis. (3 Marks)

ii. Using the given graph paper, draw the triangle *ABC* with respect to the translated axis. (1 Mark)



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(ATTEMPT EITHER PART a OR PART b OF Q.8.)

b. The *xy*-coordinate axes are rotated about the origin through an angle of $60°$. The co-ordinates of point *A* are (7, 11) with respect to *xy*-coordinate axes. Find coordinates of *A* with respect to the rotated axes. (4 Marks)

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